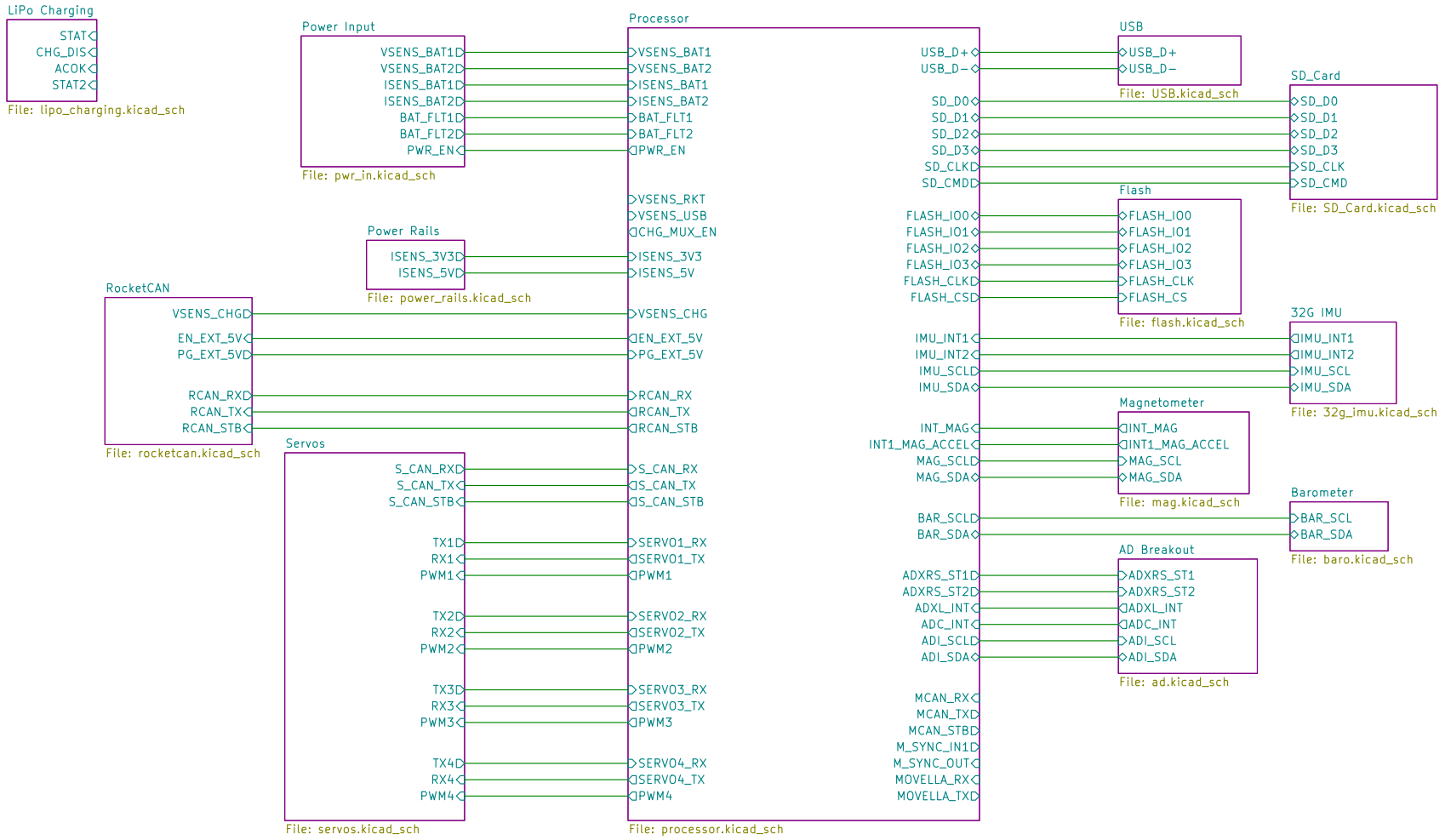
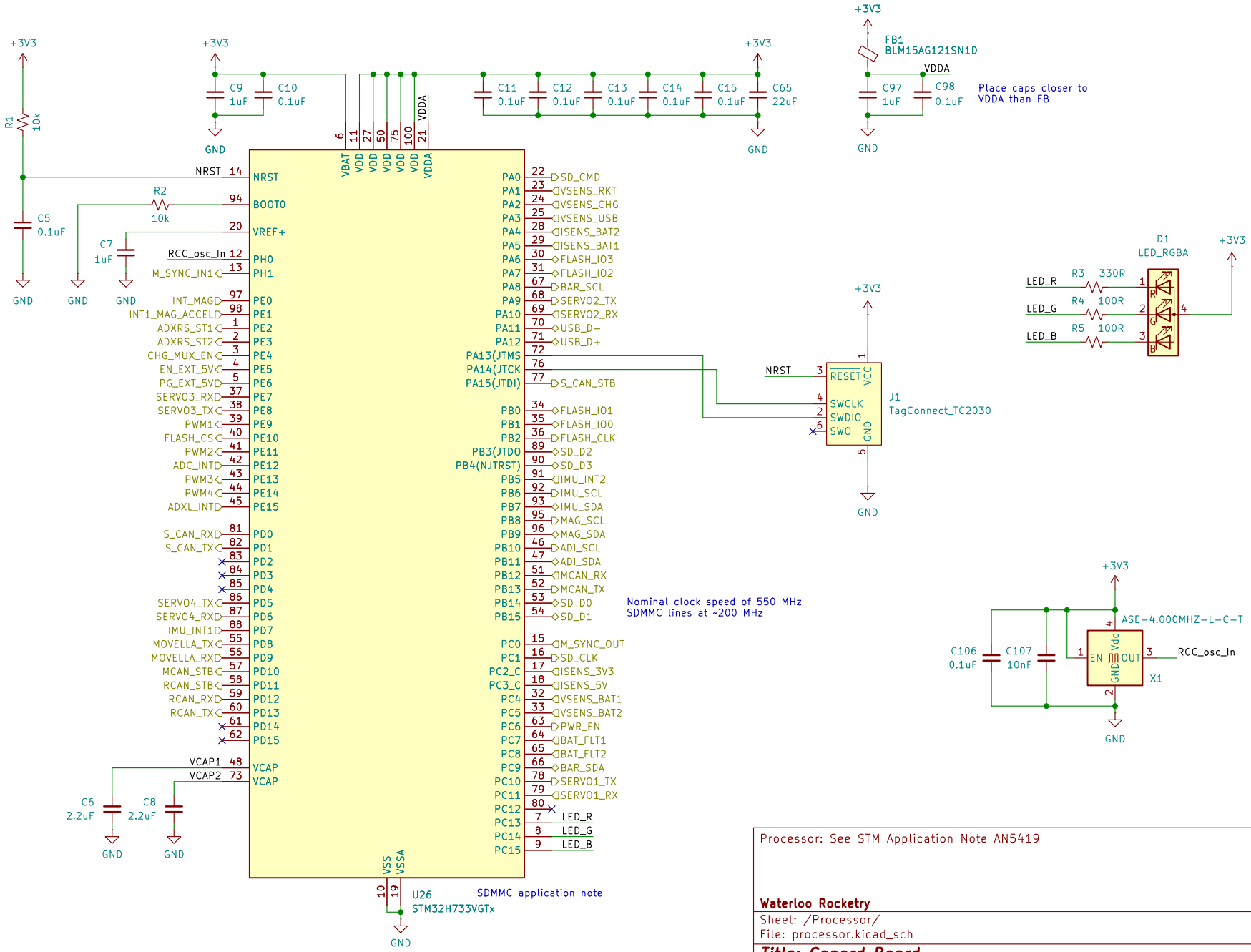
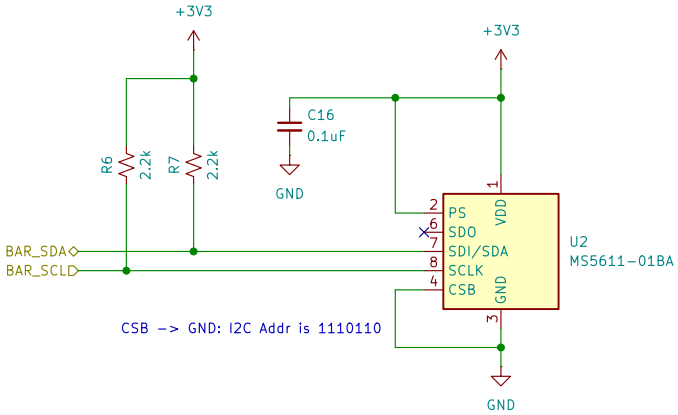

LiPo Charging
STAT
CHG_DIS
ACOK
STAT2
File: lipo_charging.kicad_sch
Power Input
VSENS_BAT1
VSENS_BAT2
ISENS_BAT1
ISENS_BAT2
BAT_FLT1
BAT_FLT2
PWR_EN
File: pwr_in.kicad_sch
Processor
VSENS_BAT1
VSENS_BAT2
ISENS_BAT1
ISENS_BAT2
BAT_FLT1
BAT_FLT2
PWR_EN
VSENS_RKT
VSENS_USB
CHG_MUX_EN
USB_D+
USB_D−
SD_D0
SD_D1
SD_D2
SD_D3
SD_CLK
SD_CMD
FLASH_IO0
FLASH_IO1
FLASH_IO2
FLASH_IO3
FLASH_CLK
FLASH_CS
IMU_INT1
IMU_INT2
IMU_SCL
IMU_SDA
INT_MAG
INT1_MAG_ACCEL
MAG_SCL
MAG_SDA
BAR_SCL
BAR_SDA
ADXRS_ST1
ADXRS_ST2
ADXL_INT
ADC_INT
ADI_SCL
ADI_SDA
MCAN_RX
MCAN_TX
MCAN_STB
M_SYNC_IN1
M_SYNC_OUT
MOVELLA_RX
MOVELLA_TX
File: processor.kicad_sch
USB
USB_D+
USB_D−
File: USB.kicad_sch
SD_Card
SD_D0
SD_D1
SD_D2
SD_D3
SD_CLK
SD_CMD
File: SD_Card.kicad_sch
Flash
FLASH_IO0
FLASH_IO1
FLASH_IO2
FLASH_IO3
FLASH_CLK
FLASH_CS
File: flash.kicad_sch
32G IMU
IMU_INT1
IMU_INT2
IMU_SCL
IMU_SDA
File: 32g_imu.kicad_sch
Magnetometer
INT_MAG
INT1_MAG_ACCEL
MAG_SCL
MAG_SDA
File: mag.kicad_sch
Barometer
BAR_SCL
BAR_SDA
File: baro.kicad_sch
AD Breakout
ADXRS_ST1
ADXRS_ST2
ADXL_INT
ADC_INT
ADI_SCL
ADI_SDA
File: ad.kicad_sch
Power Rails
ISENS_3V3
ISENS_5V
File: power_rails.kicad_sch
RocketCAN
VSENS_CHG
EN_EXT_5V
PG_EXT_5V
RCAN_RX
RCAN_TX
RCAN_STB
File: rocketcan.kicad_sch
Servos
S_CAN_RX
S_CAN_TX
S_CAN_STB
TX1
RX1
PWM1
TX2
RX2
PWM2
TX3
RX3
PWM3
TX4
RX4
PWM4
SERVO1_RX
SERVO1_TX
PWM1
SERVO2_RX
SERVO2_TX
PWM2
SERVO3_RX
SERVO3_TX
PWM3
SERVO4_RX
SERVO4_TX
PWM4
File: servos.kicad_sch
TP13
GND
H1 M3
H2 M3
H3 M3
H4 M3
H5 M3
H6 M3
SHIELD
R76 1M
C90 0.1uF
Processor: See STM Application Note AN5419
Waterloo Rocketry
Sheet: /
File: CanardBoard.kicad_sch
Title: Canard Board
Size: A4
Date: 2026-03-19
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/14


+3V3
R1
10k
C9
1uF
C10
0.1uF
GND
VBAT
VDD
VDD
VDD
VDD
VDD
VDDA
6
11
27
50
75
100
21
C11
0.1uF
C12
0.1uF
C13
0.1uF
C14
0.1uF
C15
0.1uF
C65
22uF
FB1
BLM15AG121SN1D
VDDA
C97
1uF
C98
0.1uF
Place caps closer to
VDDA than FB
NRST 14
R2
10k
94
BOOT0
20
VREF+
C5
0.1uF
C7
1uF
RCC_osc_In 12
M_SYNC_IN1 13
INT_MAGD 97
INT1_MAG_ACCELD 98
ADXRS_ST1 1
ADXRS_ST2 2
CHG_MUX_EN 3
EN_EXT_5V 4
PG_EXT_5VD 5
SERVO3_RXD 37
SERVO3_TX 38
PWM1 39
FLASH_CS 40
PWM2 41
ADC_INTD 42
PWM3 43
PWM4 44
ADXL_INTD 45
S_CAN_RXD 81
S_CAN_TX 82
83
84
85
SERVO4_TX 86
SERVO4_RXD 87
IMU_INT1D 88
MOVELLA_TX 55
MOVELLA_RXD 56
MCAN_STB 57
RCAN_STB 58
RCAN_RXD 59
RCAN_TX 60
61
62
VCAP1 48
VCAP2 73
C6
2.2uF
C8
2.2uF
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VCAP
VCAP
PA0 22
PA1 23
PA2 24
PA3 25
PA4 28
PA5 29
PA6 30
PA7 31
PA8 67
PA9 68
PA10 69
PA11 70
PA12 71
PA13(JTMS 72
PA14(JTCK 76
PA15(JTDI) 77
PB0 34
PB1 35
PB2 36
PB3(JTDO 89
PB4(NJTRST) 90
PB5 91
PB6 92
PB7 93
PB8 95
PB9 96
PB10 46
PB11 47
PB12 51
PB13 52
PB14 53
PB15 54
PC0 15
PC1 16
PC2_C 17
PC3_C 18
PC4 32
PC5 33
PC6 63
PC7 64
PC8 65
PC9 66
PC10 78
PC11 79
PC12 80
PC13 7
PC14 8
PC15 9
PA0 SD_CMD
PA1 VSENS_RKT
PA2 VSENS_CHG
PA3 VSENS_USB
PA4 ISENS_BAT2
PA5 ISENS_BAT1
PA6 FLASH_IO3
PA7 FLASH_IO2
PA8 BAR_SCL
PA9 SERVO2_TX
PA10 SERVO2_RX
PA11 USB_D−
PA12 USB_D+
PA13(JTMS SWDIO
PA14(JTCK SWCLK
PA15(JTDI) S_CAN_STB
PB0 FLASH_IO1
PB1 FLASH_IO0
PB2 FLASH_CLK
PB3(JTDO SD_D2
PB4(NJTRST) SD_D3
PB5 IMU_INT2
PB6 IMU_SCL
PB7 IMU_SDA
PB8 MAG_SCL
PB9 MAG_SDA
PB10 ADI_SCL
PB11 ADI_SDA
PB12 MCAN_RX
PB13 MCAN_TX
PB14 SD_D0
PB15 SD_D1
PC0 M_SYNC_OUT
PC1 SD_CLK
PC2_C ISENS_3V3
PC3_C ISENS_5V
PC4 VSENS_BAT1
PC5 VSENS_BAT2
PC6 PWR_EN
PC7 BAT_FLT1
PC8 BAT_FLT2
PC9 BAR_SDA
PC10 SERVO1_TX
PC11 SERVO1_RX
PC12
PC13 LED_R
PC14 LED_G
PC15 LED_B
Nominal clock speed of 550 MHz
SDMMC lines at ~200 MHz
VSS
VSSA
10
19
U26
STM32H733VGTx
SDMMC application note
NRST
RESET
VCC
SWCLK
SWDIO
SWO
GND
J1
TagConnect_TC2030
D1
LED_RGBA
LED_R
LED_G
LED_B
R3 330R
R4 100R
R5 100R
C106
0.1uF
C107
10nF
ASE−4.000MHZ−L−C−T
EN
Vdd
GND
OUT
RCC_osc_In
X1
Processor: See STM Application Note AN5419
Waterloo Rocketry
Sheet: /Processor/
File: processor.kicad_sch
Title: Canard Board
Size: A4
Date: 2026−03−19
Rev: A
KiCad E.D.A. 10.0.7
Id: 2/14



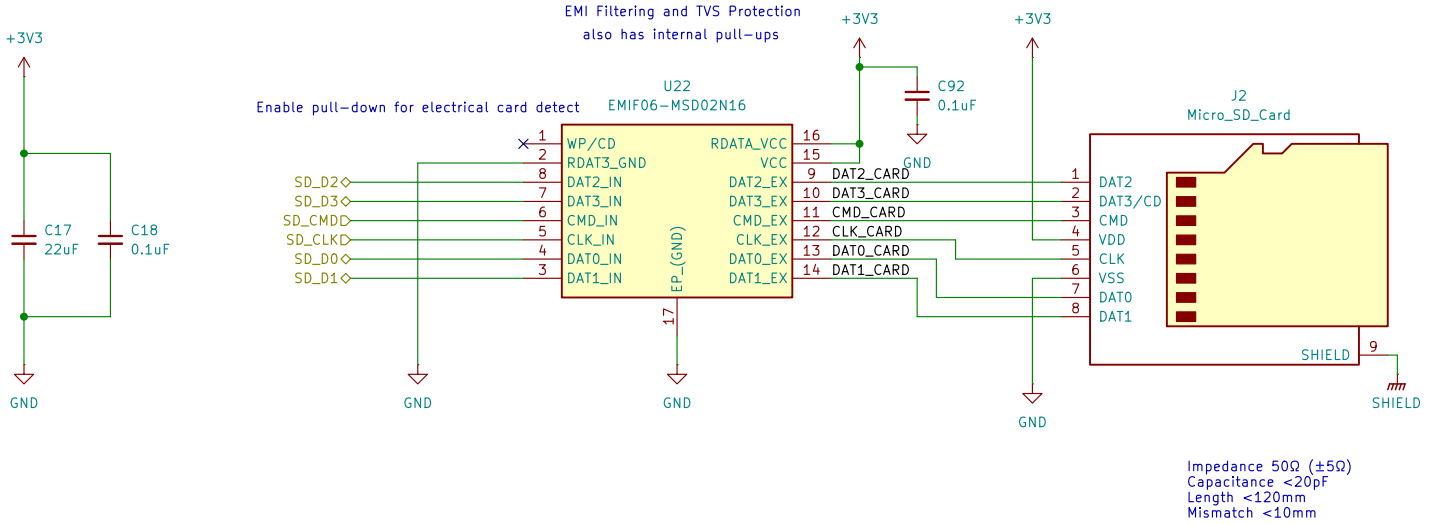

Processor: See STM Application Note AN5419

**Waterloo Rocketry**

Sheet: /Barometer/
File: baro.kicad_sch

***Title: Canard Board***

| Size: A4 | Date: 2026-03-19 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/14 |



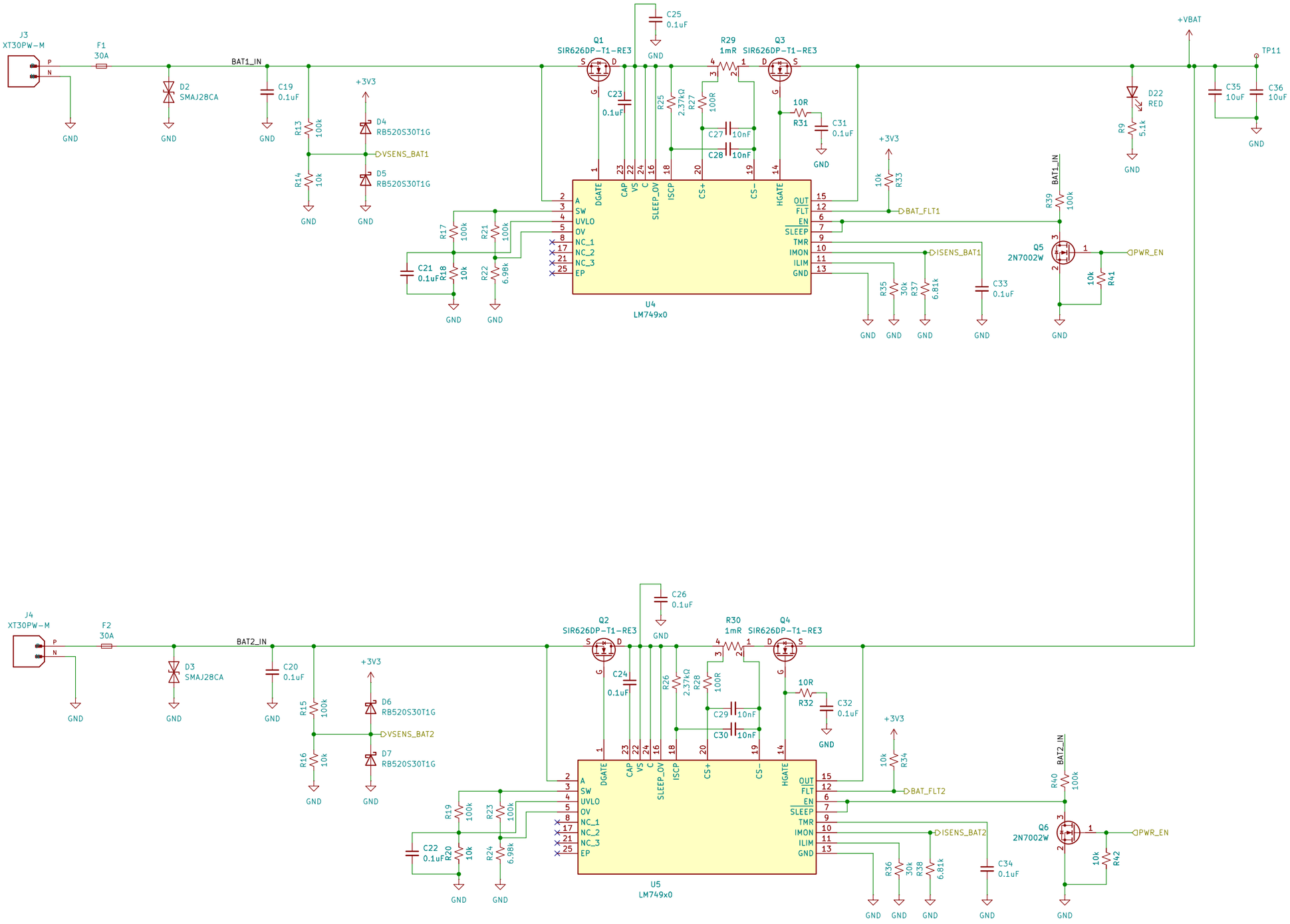

Processor: See STM Application Note AN5419

**Waterloo Rocketry**

Sheet: /SD_Card/
File: SD_Card.kicad_sch

***Title: Canard Board***

| Size: A4 | Date: 2026-03-19 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/14 |



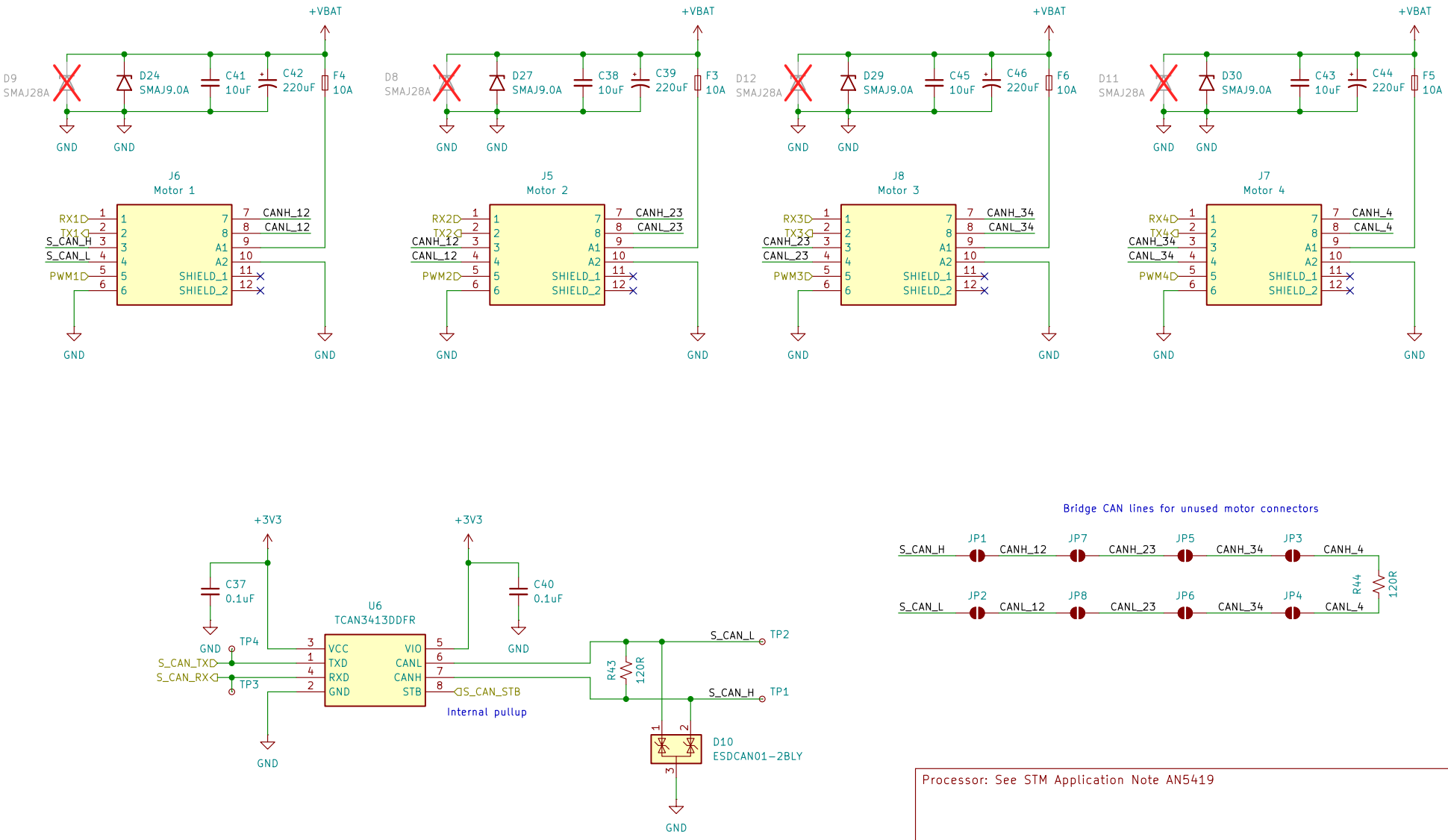

Processor: See STM Application Note AN5419

**Waterloo Rocketry**

Sheet: /Power Input/
File: pwr_in.kicad_sch

***Title: Canard Board***

| Size: A3 | Date: 2026-03-19 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/14 |



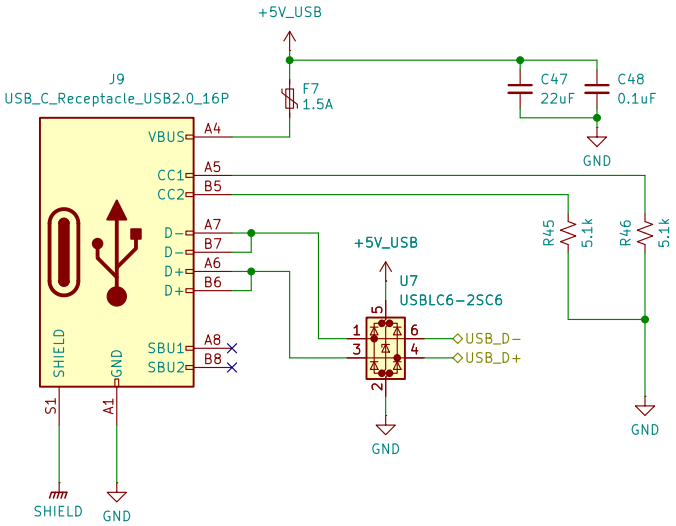

Processor: See STM Application Note AN5419

**Waterloo Rocketry**

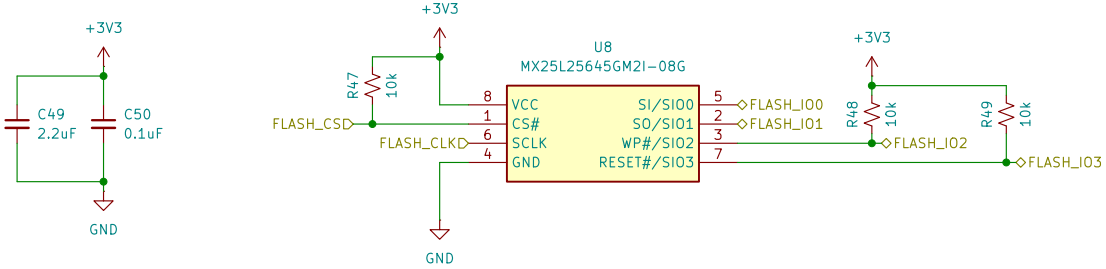
Sheet: /Servos/
File: servos.kicad_sch

***Title: Canard Board***

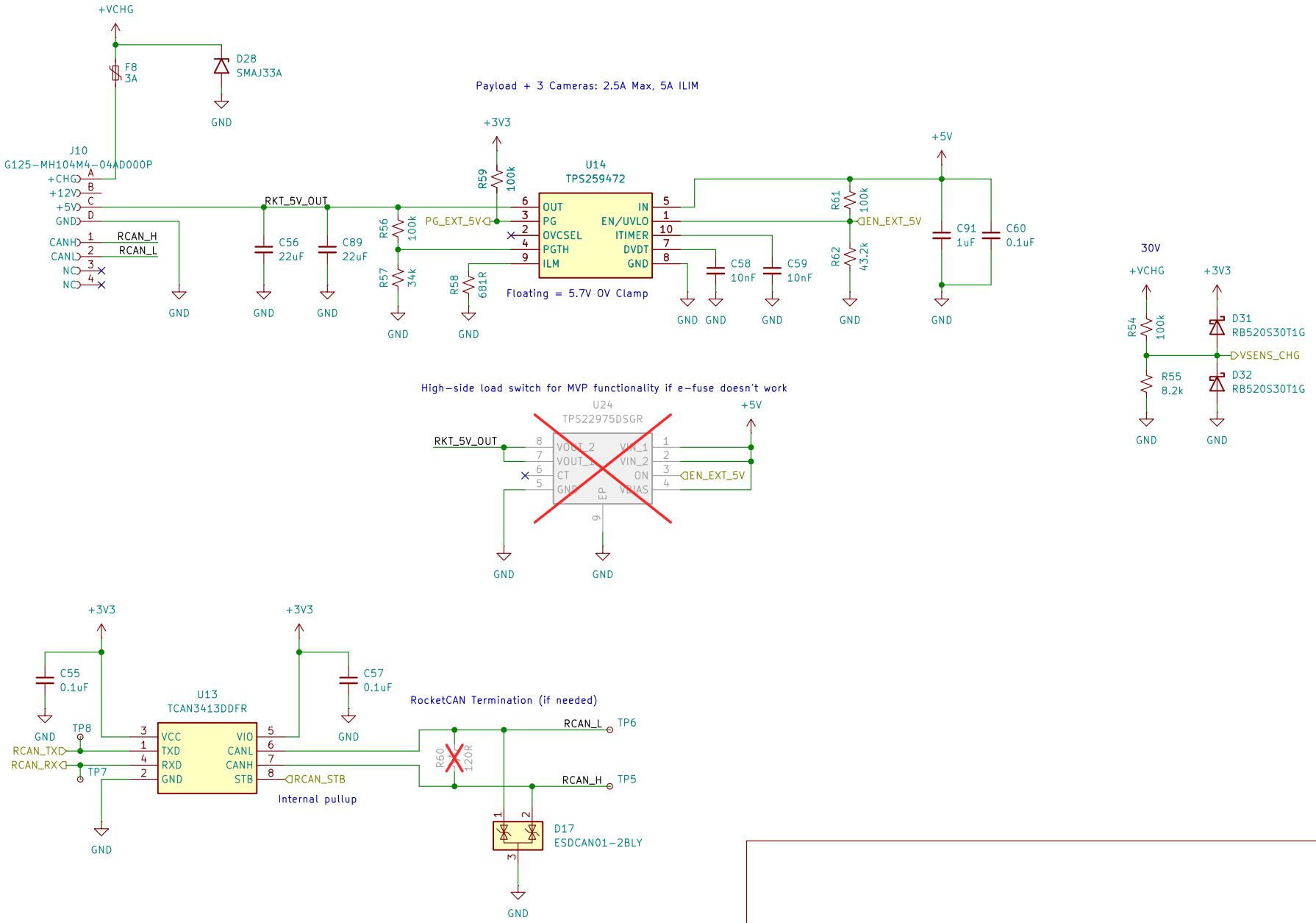
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/14 |



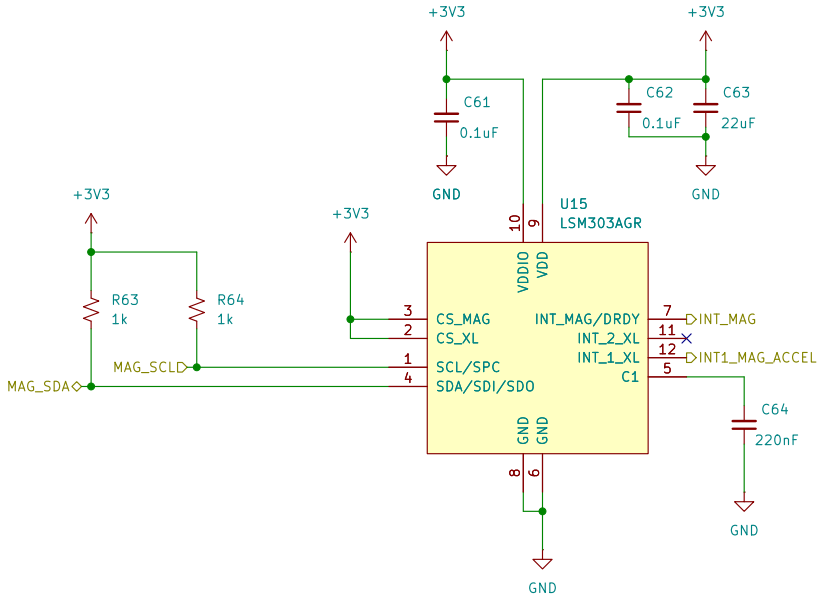

| | | |
|---|---|---|
| Sheet: /USB/ | | |
| File: USB.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 7/14 |




| | | |
|---|---|---|
| Sheet: /Flash/ | | |
| File: flash.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 8/14 |



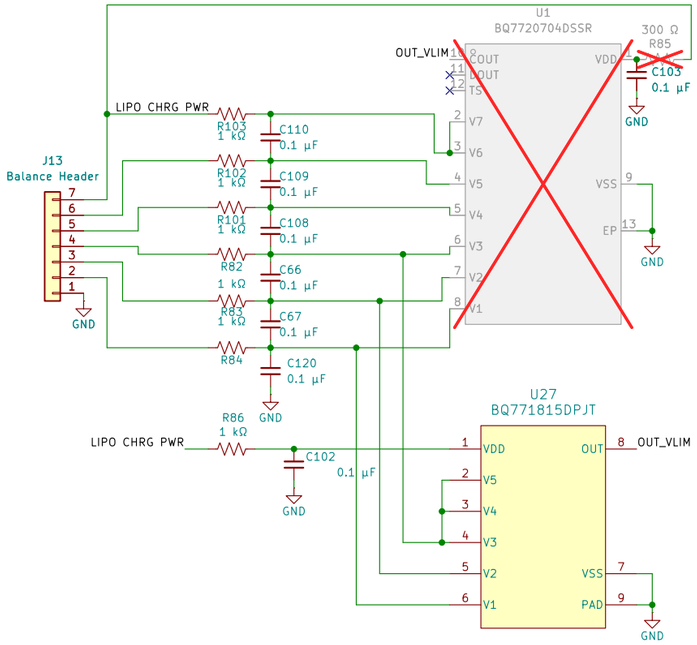

| | | |
|---|---|---|
| Sheet: /RocketCAN/ | | |
| File: rocketcan.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 10/14 |



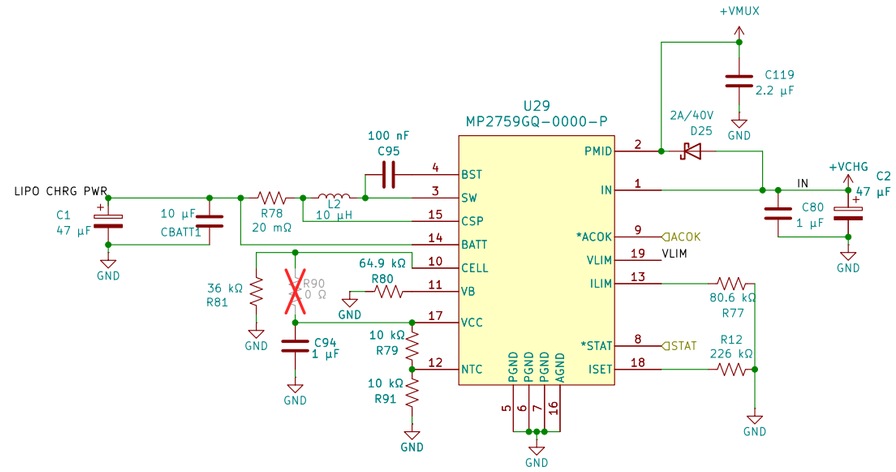

IIS2MDC also works, just doesn't have the accel

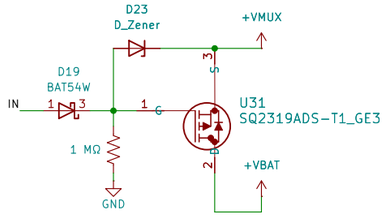
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|---|---|---|
| Sheet: /Magnetometer/<br>File: mag.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 11/14 |

We need to check which way is which for the lipo

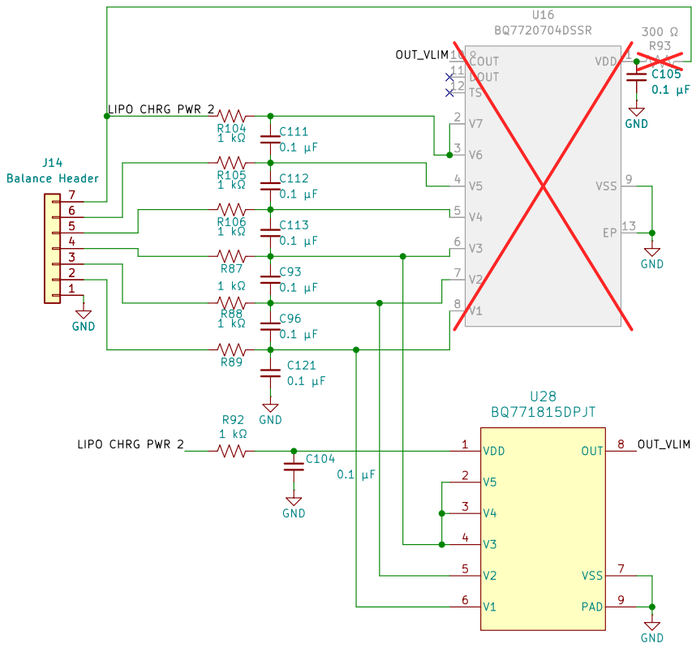



Resistor Disconnected so IC is unused for 2S

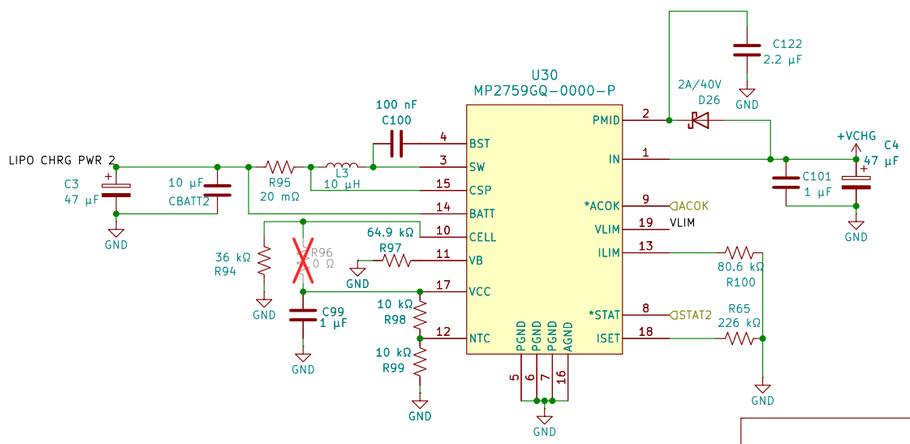


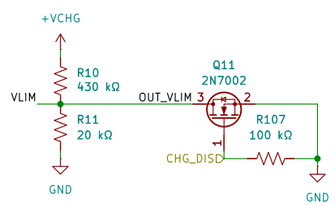




We need to check which way is which for the lipo




Resistor Disconnected so IC is unused for 2S

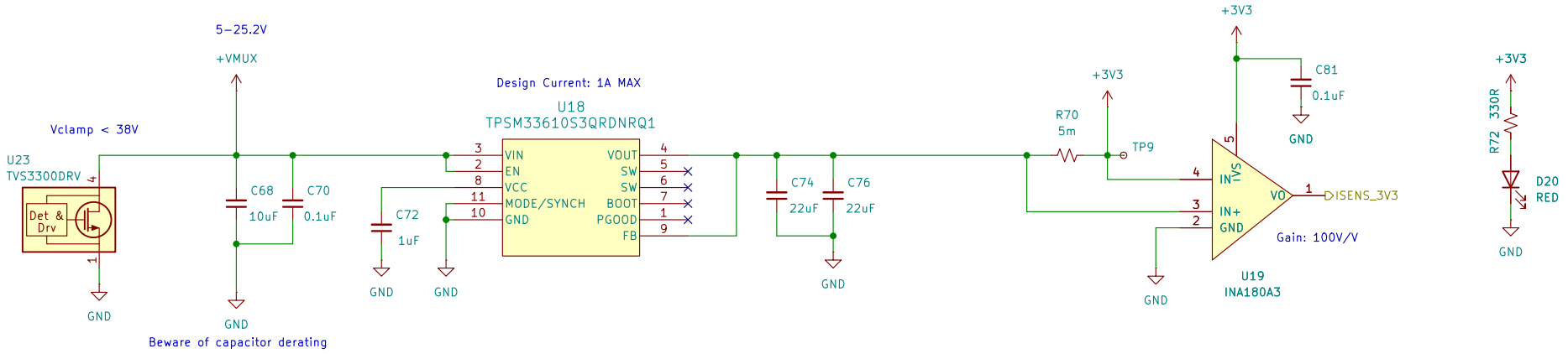


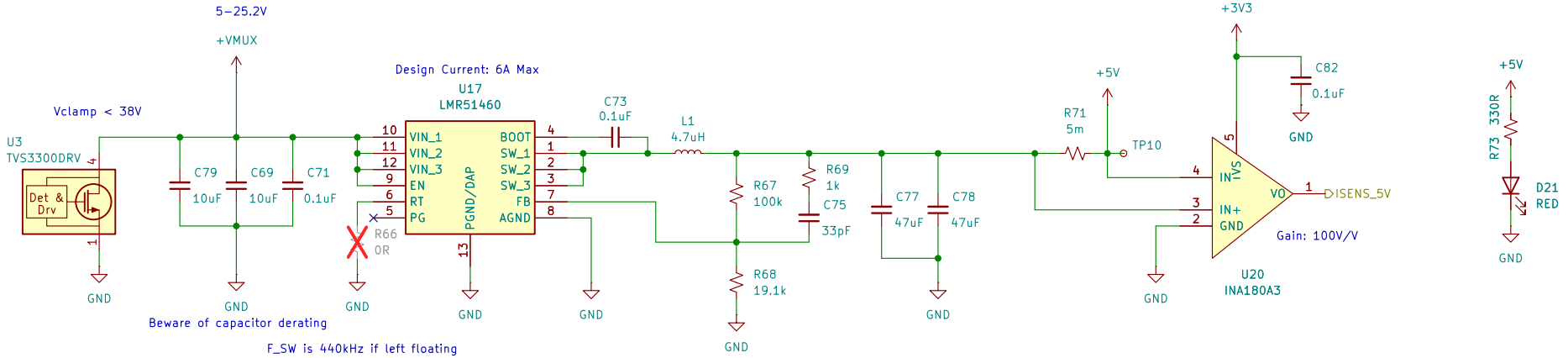




Sheet: /LiPo Charging/
File: lipo_charging.kicad_sch

**Title:**

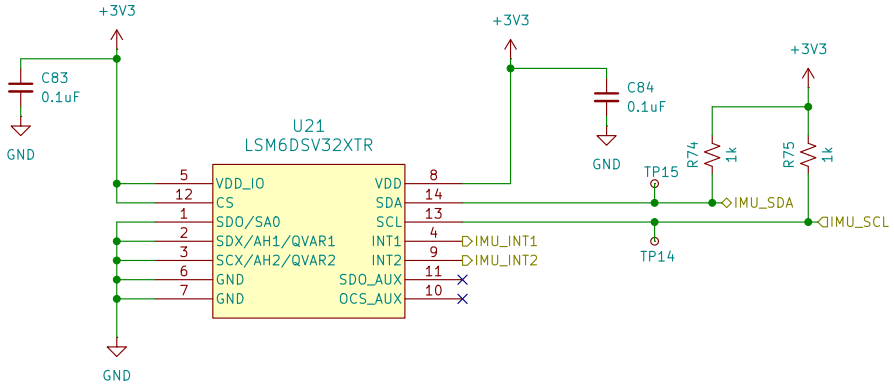
| Size: A3 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 12/14 |



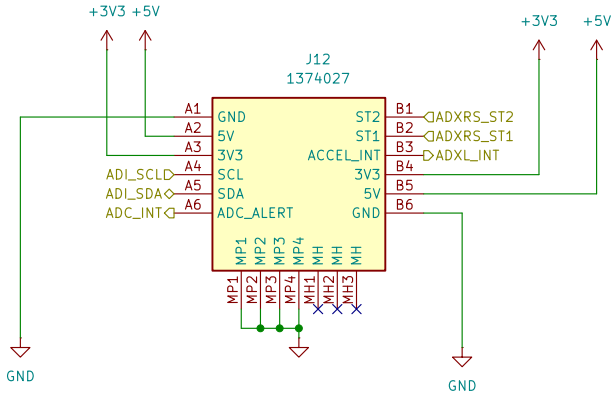




https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=BC8C0EE24EA233AC

| | | |
|---|---|---|
| Sheet: /Power Rails/ | | |
| File: power_rails.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 13/14 |




| | | |
|---|---|---|
| Sheet: /32G IMU/ | | |
| File: 32g_imu.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 14/14 |




I2C pullups on breakout board

| | | |
|---|---|---|
| Sheet: /AD Breakout/ | | |
| File: ad.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-03-19 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 15/14 |